

The height of filling is determined by a single gage point instead of the three points of the Engler and the overflow rim of the Saybolt. The instrument is provided with leveling screws.⁶

3. DIMENSIONS

The essential dimensions of the Redwood viscometer are given in Table 1, according to different experimenters.

TABLE 1.—Dimensions of Redwood Viscometer

Name of dimension	Higgins ^a		Meissner ^b No. 1263	Bureau of Standards	
	Normal	Tolerance		No. 1004	No. 1025
	cm	cm ±	cm	cm	cm
Diameter of outlet tube <i>d</i>	0.15	0.003	0.158	^c 0.1654	0.1583
Length of outlet tube, <i>l</i>	1.00	.01	1.023	.9853	.9639
Height of gage point above bottom of outlet tube, <i>h</i> ₁ ..	9.25	<i>d</i> .64	9.615	8.861	9.400
Diameter of container, <i>D</i>	4.68	.02	4.658	4.589	4.695
Average head (calculated) <i>h</i>	^e 7.67		8.059	7.211	7.838
Diameter of stem of oil-cup thermometer.....	.63				
Diameter of bulb of oil-cup thermometer.....	.80				
External diameter of agate, upper end <i>f</i>	1.27	.10			
External diameter of agate, lower end <i>f</i>	1.15	.10			
Height of base of container <i>g</i>	.80	.05	.852		

^a W. F. Higgins, Collected researches, National Physical Laboratory, 11, p. 5; 1914.
^b W. Meissner, Chem. revue über die fett und harz industrie, 19, p. 9; 1912.
^c This is considerably above the normal dimension reported by Higgins, but Archbutt and Deeley, p. 166, give *d* as "very nearly" 0.17 cm, with *l*=1.2 cm.
^d Higgins gives the initial head in two parts, the height of gage point above the upper end of outlet tube being 8.25±0.63.
^e Not given by Higgins.
^f The importance of these dimensions appears to be mainly in shop construction.
^g This is the height of junction of cylindrical portion of oil cup with concave bottom of cup above upper end of outlet tube.

The last dimension of Table 1 appears to be needlessly restricted. As 50 cc of oil are run out in a test, the oil level falls 2.96 cm, so that this dimension could have a maximum value of 5.29 cm (instead of 0.85), and the oil surface at the end of run would still be within the cylindrical part of the container, which is a necessary condition in order that the average head may be correct.

The bulb of the thermometer must be placed somewhere within the above-mentioned 2.96 cm, so as not to be exposed at the end of run and so as not to be too near the outlet tube and thus obstruct the flow. It seems probable that more reproducible results could be obtained if the exact location were specified.

There is some uncertainty about the standard dimensions, because Higgins says he expressed them in millimeters for conven-

⁶ For a consideration of the error due to inaccuracy in leveling see W. F. Higgins, Collected researches, National Physical Laboratory, 11, p. 6; 1914.